

Run Wang

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SUMMARY

Computer Science Ph.D. student specializing in 3D computer vision, AI security, and mobile/edge computing, with experience spanning research, ML systems, and production software engineering. Published/accepted work in CVPR, IEEE INFOCOM, and IEEE/ACM IWQoS.

SKILLS

Programming: C, C++, C#, Python (PyTorch), Java

Cloud / DevOps: Docker, Kubernetes, Terraform

Research Areas: 3D Computer Vision, AI Security, Autonomous Driving, Mobile Computing

EDUCATION

Clemson University

Clemson, SC

Ph.D. Student in Computer Science

Aug. 2025 – Present

– Concentration: 3D Computer Vision, AI Security and Mobile Computing

Carnegie Mellon University

Pittsburgh, PA

M.S. in Electrical and Computer Engineering (Applied Advanced Program)

Sep. 2021 – May 2023

Rensselaer Polytechnic Institute

Troy, NY

B.S. in Computer & Systems Engineering

Sep. 2017 – May 2021

PUBLICATIONS AND PREPRINTS

R. Wang et al., “FlexMap: Generalized HD Map Construction from Flexible Camera Configurations”
arXiv Preprint, 2026.

C. Zhou, **R. Wang et al.**, “FF3R: Feedforward Feature 3D Reconstruction from Unconstrained Views”
CVPR, 2026.

Q. Yang, C. Zhou, X. Liu, **R. Wang et al.**, “Sat3R: Satellite DSM Reconstruction via RPC-Aware Depth Fine-tuning”
CVPRW, 2026.

R. Wang et al., “Lightweight Detection of Abnormal Battery Drain Induced by Network Operations of Mobile Apps”
IEEE INFOCOM, 2026. Acceptance Rate: 329/1740 (18.91%)

R. Wang et al., “MAPP: Predictive UI View Pre-caching for Improving the Responsiveness of Mobile Apps”
IEEE/ACM IWQoS, 2025. Acceptance Rate: 81/326 (24.8%)

Z. Wang, M. Kim, **R. Wang et al.**, “LLM Agents for Simulating Human Stated-Choice Behavior”
ICMLA, 2026.

MANUSCRIPTS CURRENTLY UNDER REVIEW

R. Wang et al., “Adversarial Prompts for Acceptance Collapse in Speculative Decoding”

X. Liu, C. Zhou, **R. Wang et al.**, “Lang-SVG: Hierarchical Image Vectorization with Language Priors”

Y. Liu, **R. Wang et al.**, “Who Touched the Screen? Driver-Passenger Identification in In-Vehicle Infotainment (IVI) Systems”

P. MohajerAnsari, A. Salarpour, **R. Wang et al.**, “GATE: Reliability-Gated Gaussian Evidence Fusion for Training-Free Test-Time Adaptation of Vision-Language Models”

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2025 – Present

Clemson University

Clemson, SC

- Supervisors: [Dr. Siyu Huang](#) and [Dr. Mert D. Pesé](#)
- Ongoing Project: Researching AI security and robustness of autonomous-driving systems, focusing on adversarial vulnerabilities in perception and decision-making pipelines.

Graduate Research Associate

May 2024 – Aug. 2025

The Ohio State University

Columbus, OH

- Supervisor: Prof. Xiaorui Wang
- Designed ML and time-series models for mobile/edge performance optimization, producing two peer-reviewed papers accepted to IEEE/ACM IWQoS 2025 and IEEE INFOCOM 2026.

Research Assistant

May 2022 – May 2023

CERLAB, Carnegie Mellon University

Pittsburgh, PA

- Supervisors: Prof. Kenji Shimada and Dr. Wentai Zhang
- Adapted PSPNet, DeepLabV3+, and SketchGNN for component extraction from engineering drawings, achieving 84% overall accuracy on in-house dataset. (acknowledged by this paper: [Component segmentation of engineering drawings using Graph Convolutional Networks](#))

Research Assistant

Jan. 2022 – May 2022

CyLab, Carnegie Mellon University

Pittsburgh, PA

- Supervisor: Dr. Cai Yang
- Collected and dynamically traced 80+ Linux malware samples, extracting 14 system-level behavioral features for malware behavior analysis.
- Built an in-house behavioral dataset from runtime traces to support downstream malware classification and anomaly detection experiments.

Research Assistant

Jan. 2020 – Dec. 2020

RPI-IBM Cognitive and Immersive Systems Lab

Troy, NY

- Supervisors: Dr. Jeff Kephart and Dr. Xiangyang Mou
- Built real-time gesture recognition system in C# and Python using Intel RealSense cameras, collecting 600+ gesture samples and extracting skeletal motion features for recognition experiments.

INDUSTRY EXPERIENCE

Application Developer – Full Stack

May 2023 – Apr. 2024

Computer Packages Inc.

Rockville, MD

- Developed and maintained C#/.NET, JavaScript, and SQL Server features for legacy-to-modern application migration, preserving data integrity across production workflows.
- Optimized SQL Server query performance using stored procedures, indexing, and execution-plan analysis, reducing query latency by > 50%.

PROJECT EXPERIENCE

Cosmospapers Research Assistant Platform

Jan. 2026 – Present

[Demo](#) - [Github](#)

- Built RAG-based research discovery platform indexing 54,000+ papers from 15+ top CS conferences with vectorized retrieval and interactive topic exploration.
- Extracted 150 hierarchical research topics and visualized topic relationships to help users explore papers by research area.
- Shipped public demo and GitHub repository, attracting 23 stars.

User Recommendation Microservice

Aug. 2022 – Dec. 2022

GCP, Kubernetes, Spark, FastAPI, Vert.x

- Built Spark ETL pipelines on GCP Dataproc to preprocess $\approx 1TB$ Twitter dataset for recommendation-system experiments.
- Deployed cloud-native recommendation microservices using Kubernetes, FastAPI, and Vert.x, achieving 10,000 RPS within a \$0.70/hour infrastructure budget.